GenOptics: An intuitive platform of visual analytics for integrative analysis of large-scale multi-omics data

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Highlights

- Continuously increasing volume of large-scale multi-omics data
- The increasing availability of free and commercial software tools along with the need for integrative analysis of multi-omics data, are beset by a number of problems which include
  - Software incompatibility
  - Complex parameter selection
  - Shortage of trained personnel
- GenOptics is a visual analytics platform that aims to facilitate the integration and subsequent analysis of diverse multi-omics datasets
- The platform is open-source, user-friendly and offers reproducibility and full data control by the user

GenOptics platform architecture

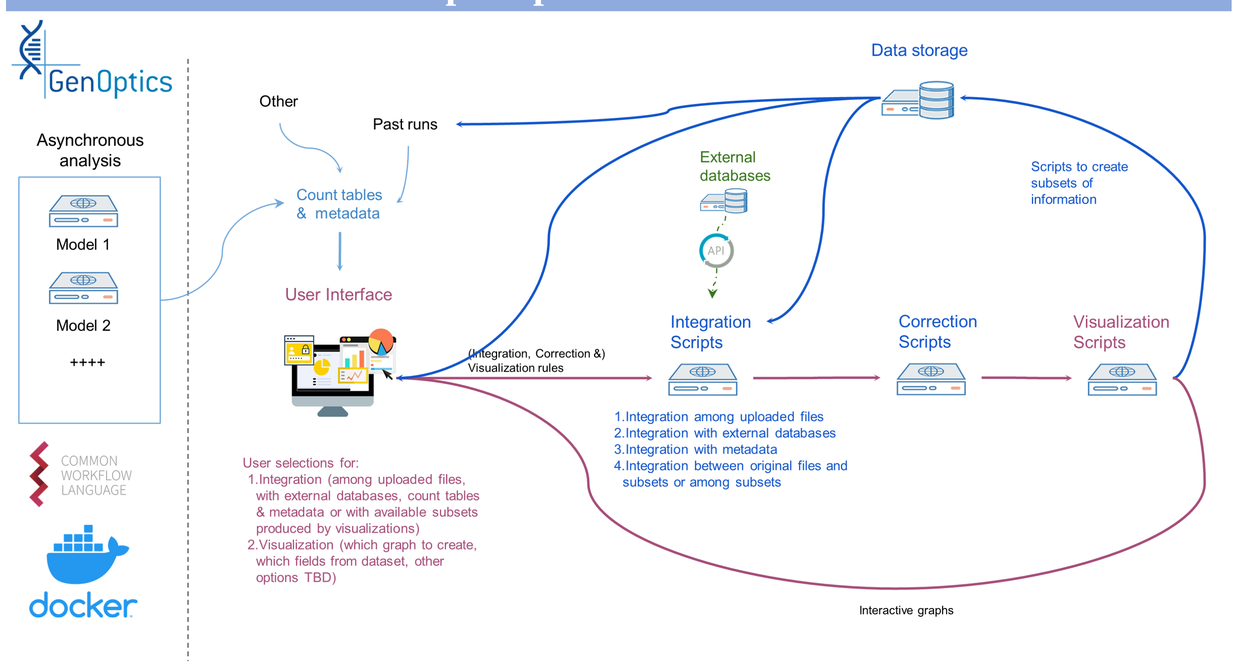

Fig 1. A schematic representation of GenOptics platform architecture. This includes the asynchronous analyses of large-scale omics data (left), implemented with CWL workflows and Docker containers, and the operations of the interactive dashboard of visual analytics (right).

CWL workflows for asynchronous analyses of large-scale omics data

The GenOptics platform offers data processing workflows for (1) RNA-Seq, (2) Whole-Exome Seq and (3) ChIP-Seq. The workflows are implemented with the use of Common Workflow Language[1] and Docker containers to automate software installation and confer cross-platform portability.

Visual analytics platform

GenOptics is heavily invested on making rich and comprehensive visualizations, and adopts several software tools, including the plotly library[2], ggplot2 and ComplexHeatmap[3]. The platform itself is a containerized application that relies on three independent containers: The Backend API container, (2) the Frontend container and (3) the PostgreSQL container.

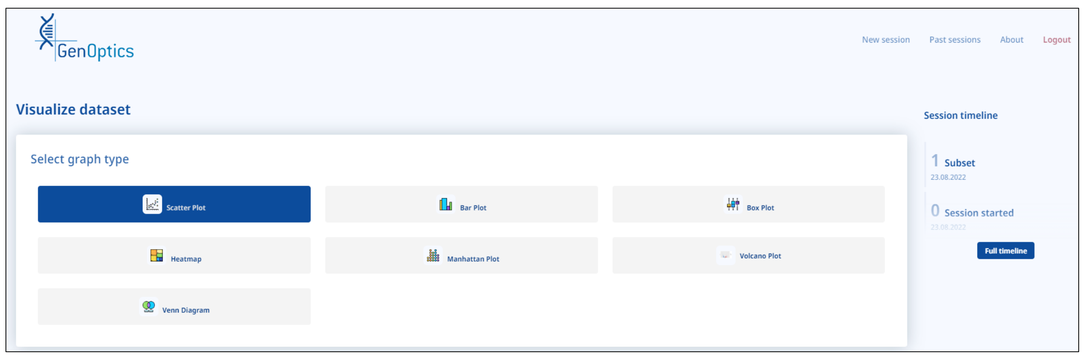

Fig 2. Table of options for data visualization through GenOptics. The available options for graph creation include: (1) Scatter plot, (2) Bar plot, (3) Box plot, (4) Heatmap, (5) Manhattan plot, (6) Volcano plot, and (7) Venn diagram.

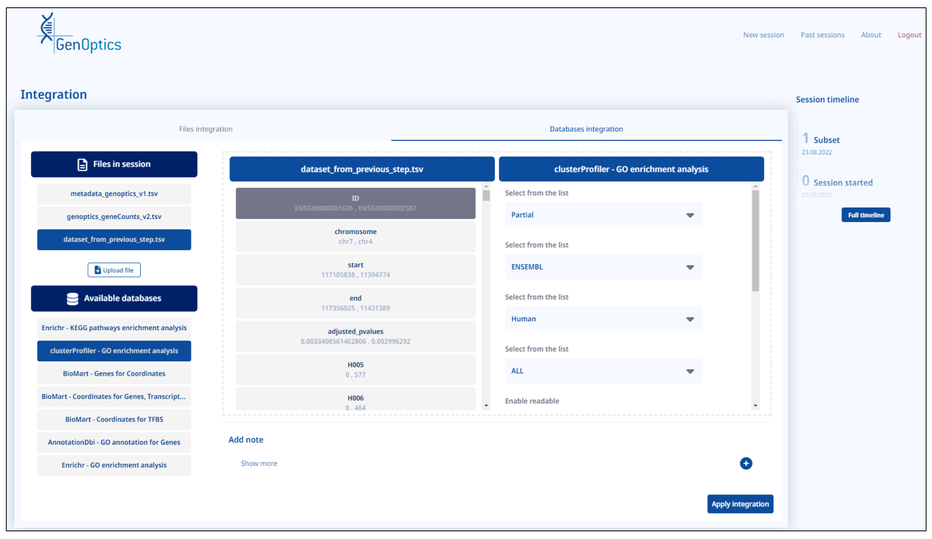

Fig 3. Integration with external database (Gene Ontology; GO) to annotate gene list of Ensemble identifiers. The annotation is achieved through the implementation of the R library clusterProfiler[4].

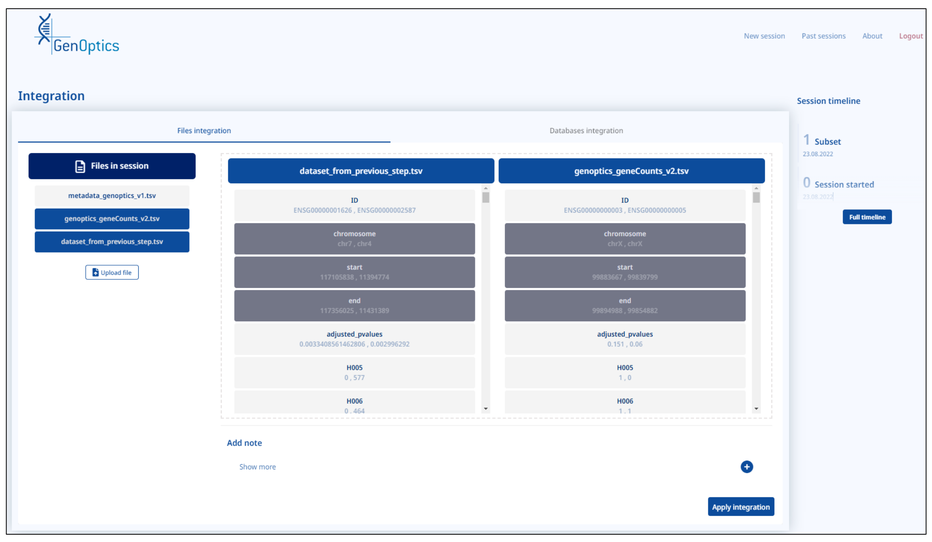

Fig 4. Integration of two different datasets based on overlapping chromosomal coordinates. The integration is achieved through the implementation of the R library GenomicRanges[5].

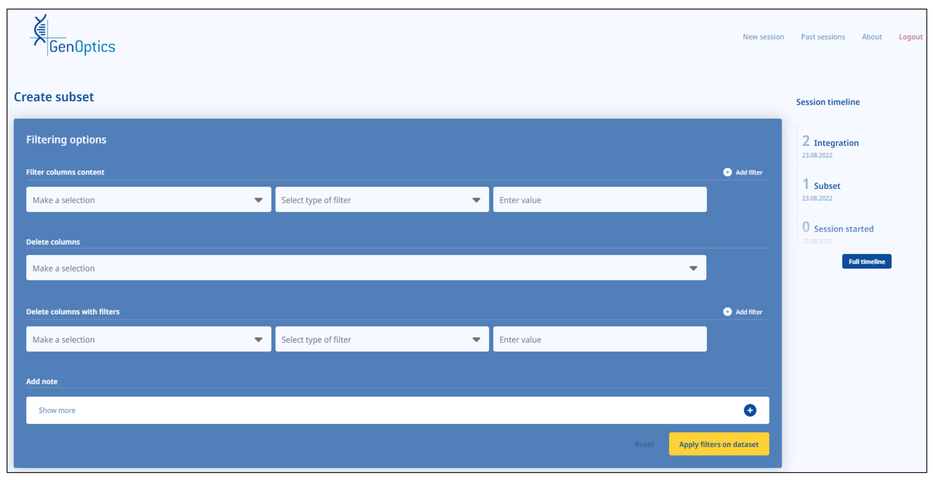

Fig 5. Table of options for subsetting and/or filtering datasets through GenOptics. The available operations include: (1) column content filtering through logical operations or column removal either (2) by selection or (3) through logical operations.

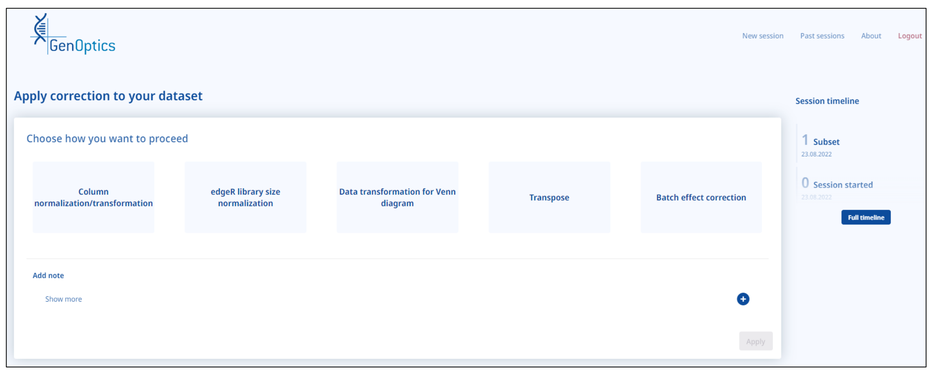

Fig 6. Table of options for applying dataset corrections through GenOptics. Available options include: (1) normalization/transformation of column(s) with numeric values, (2) data manipulation before Venn diagram plotting, (3) matrix transpose, and (4) batch effect corrections.

Acknowledgements

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